

Assignment 3

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Configure

```
In [109... %config SqlMagic.style = '_DEPRECATED_DEFAULT'
```

Load Proper Libraries

```
In [110... %load_ext sql
```

The sql extension is already loaded. To reload it, use:
%reload_ext sql

Load in Database

```
In [111... %sql sqlite:///airline_industry.db
```

Q1. How many records are in each table in the database?

Table Names

```
In [112... %%sql
select name
from sqlite_master
where type = 'table';
```

* sqlite:///airline_industry.db
Done.

```
Out[112...  name
```

planes

weather

airports

airlines

flights

Count for Planes

```
In [113... %%sql
select
    count(*)
from
    planes

* sqlite:///airline_industry.db
Done.
```

```
Out[113... count(*)
3322
```

Count for Weather

```
In [114... %%sql
select
    count(*)
from
    weather

* sqlite:///airline_industry.db
Done.
```

```
Out[114... count(*)
26115
```

Count for Airports

```
In [115... %%sql
select
    count(*)
from
    airports

* sqlite:///airline_industry.db
Done.
```

```
Out[115... count(*)
1458
```

Count for Airlines

```
In [116... %%sql
```

```
select
  count(*)
from
  airlines
```

```
* sqlite:///airline_industry.db
Done.
```

```
Out[116... count(*)
          16
```

Count for Flights

```
In [117... %%sql
select
  count(*)
from
  flights
```

```
* sqlite:///airline_industry.db
Done.
```

```
Out[117... count(*)
          336776
```

Q2. Did any flights depart and arrive exactly on time?

```
In [118... %%sql
select
  *
from
  flights
limit
  5
```

```
* sqlite:///airline_industry.db
Done.
```

Out[118...

field1	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	sch
1	2013	1	1	517	515	2	830	
2	2013	1	1	533	529	4	850	
3	2013	1	1	542	540	2	923	
4	2013	1	1	544	545	-1	1004	
5	2013	1	1	554	600	-6	812	

In [119...

```

%%sql
select count(*) as 'Flights On Time'
from
    flights
where
    dep_time = sched_dep_time
and
    arr_time = sched_arr_time

```

* sqlite:///airline_industry.db
Done.

Out[119...

Flights On Time
347

Yes, 347 flights departed and arrived on time

Q3. What is the mean arrival delay among flights?

In [120...

```

%%sql
select
    avg(arr_delay) as 'Average Arrival Delay'
from
    flights

```

* sqlite:///airline_industry.db
Done.

Out[120... **Average Arrival Delay**

6.702300639000404

Q4. How many flights had above average arrival delays?

```
In [121... %%sql
select count(*) as 'Number of Flights'
from
    flights
having arr_delay > avg(arr_delay)
```

* sqlite:///airline_industry.db
Done.

Out[121... **Number of Flights**

336776

**Q5. What is the relationship between temperature and wind speed?
Do changes in the mean temperature throughout the year (by
month) move reflect changes in mean wind speed?**

```
In [122... %%sql
select *
from
    weather
limit 5
```

* sqlite:///airline_industry.db
Done.

Out[122... **field1** **origin** **year** **month** **day** **hour** **temp** **dewp** **humid** **wind_dir** **wind_sp**

1	EWR	2013	1	1	1	39.02	26.06	59.37	270	10.35
---	-----	------	---	---	---	-------	-------	-------	-----	-------

2	EWR	2013	1	1	2	39.02	26.96	61.63	250	8.05
---	-----	------	---	---	---	-------	-------	-------	-----	------

3	EWR	2013	1	1	3	39.02	28.04	64.43	240	11.5
---	-----	------	---	---	---	-------	-------	-------	-----	------

4	EWR	2013	1	1	4	39.92	28.04	62.21	250	12.65
---	-----	------	---	---	---	-------	-------	-------	-----	-------

5	EWR	2013	1	1	5	39.02	28.04	64.43	260	12.65
---	-----	------	---	---	---	-------	-------	-------	-----	-------

In [123...

```
%%sql
select
    year,
    month,
    avg(temp),
    avg(wind_speed)
from
    weather
group by
    year,
    month
```

* sqlite:///airline_industry.db

Done.

Out [123...

year	month	avg(temp)	avg(wind_speed)
2013	1	35.63566037735849	11.18365846361186
2013	2	34.27059701492537	12.67403823880597
2013	3	39.88007184553211	12.923150884598114
2013	4	51.74564150069477	11.098444261232052
2013	5	61.795	9.516620627240142
2013	6	72.184	10.26325274074074
2013	7	80.06622082585278	9.57347724416517
2013	8	74.43487595850247	8.611903915200722
2013	9	67.37129226493747	8.906685409911997
2013	10	60.07113019891501	9.702032649186256
2013	11	44.9904343764596	11.801810583839327
2013	12	38.44180037313433	10.131050606343283

Q6. Print the first 5 rows of flights and airlines. Does the flights table contain the full name of the airline? If so, print them. If not, write a query to return the first 10 records in the flights data with the following columns: the origin, destination, carrier, flight number, and full name of the airline for each flight in a single table.

In [124...

```
%%sql
select *
from
    flights
limit 5
```

* sqlite:///airline_industry.db

Done.

Out [124...

field1	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	sch
1	2013	1	1	517	515	2	830	
2	2013	1	1	533	529	4	850	
3	2013	1	1	542	540	2	923	
4	2013	1	1	544	545	-1	1004	
5	2013	1	1	554	600	-6	812	

In [125...

```
%%sql
select *
from
    airlines
limit 5
```

* sqlite:///airline_industry.db
Done.

Out [125...

field1	carrier	name
1	9E	Endeavor Air Inc.
2	AA	American Airlines Inc.
3	AS	Alaska Airlines Inc.
4	B6	JetBlue Airways
5	DL	Delta Air Lines Inc.

No, the flight table does not contain the full name of the airline

In [126...

```
%%sql
select
    f.origin,
    f.dest,
    f.carrier,
    f.flight,
    a.name
from
```

```

    flights f
left join airlines a
    on f.carrier = a.carrier
limit 10

```

```
* sqlite:///airline_industry.db
```

Done.

Out [126...

origin	dest	carrier	flight	name
EWR	IAH	UA	1545	United Air Lines Inc.
LGA	IAH	UA	1714	United Air Lines Inc.
JFK	MIA	AA	1141	American Airlines Inc.
JFK	BQN	B6	725	JetBlue Airways
LGA	ATL	DL	461	Delta Air Lines Inc.
EWR	ORD	UA	1696	United Air Lines Inc.
EWR	FLL	B6	507	JetBlue Airways
LGA	IAD	EV	5708	ExpressJet Airlines Inc.
JFK	MCO	B6	79	JetBlue Airways
LGA	ORD	AA	301	American Airlines Inc.

Q7. Your supervisor is interested in the airlines that offer the most flights in peak periods. Write a query that returns the number of flights each airline operates in December. The results should be a single table that contains five columns: carrier, month, name, and num_flights. The results should be limited to 10 and shown in descending order by number of flights.

In [127...

```

%%sql
select
    f.carrier,
    month,
    name,
    count(*) as num_flights
from
    flights f
left join airlines a
    on f.carrier = a.carrier
where
    month = 12
group by
    f.carrier,
    month,
    name
order by
    num_flights DESC

```



```
limit 10
```

```
* sqlite:///airline_industry.db
```

Done.

```
Out [127... carrier month name num_flights
```

carrier	month	name	num_flights
UA	12	United Air Lines Inc.	4931
B6	12	JetBlue Airways	4741
EV	12	ExpressJet Airlines Inc.	4307
DL	12	Delta Air Lines Inc.	4093
AA	12	American Airlines Inc.	2705
MQ	12	Envoy Air	2139
9E	12	Endeavor Air Inc.	1633
US	12	US Airways Inc.	1605
WN	12	Southwest Airlines Co.	1099
VX	12	Virgin America	476

Q8. Use pandas to connect to the database. Define a function that takes one argument--month_num--called num_flights_fun that uses pandas' `.read_sql_query()` and runs a query that returns the number of flights each airline operates in whatever month is specified in the function call. The result should be returned as a pandas dataframe that contains five columns: carrier, month, name, and num_flights.

```
In [128... import pandas as pd
import sqlite3

conn = sqlite3.connect("airline_industry.db")
```

```
In [129... def num_flights_fun(month_num):
    """
    month_num : int
        Month number (1-12)

    pandas.DataFrame
        Columns: carrier, month, name, num_flights
    """
    query = """
        select
            f.carrier,
            f.month,
            a.name,
            count(*) AS num_flights
        from
```

```

        flights f
    join
        airlines a
    on f.carrier = a.carrier
    where
        f.month = ?
    group by
        f.carrier, f.month, a.name
    order by
        num_flights DESC;
"""

df = pd.read_sql_query(query, conn, params=(month_num,))
return df

```

In [130... num_flights_fun(7)

Out[130...

	carrier	month	name	num_flights
0	UA	7	United Air Lines Inc.	5066
1	B6	7	JetBlue Airways	4984
2	EV	7	ExpressJet Airlines Inc.	4641
3	DL	7	Delta Air Lines Inc.	4251
4	AA	7	American Airlines Inc.	2882
5	MQ	7	Envoy Air	2261
6	US	7	US Airways Inc.	1786
7	9E	7	Endeavor Air Inc.	1494
8	WN	7	Southwest Airlines Co.	1076
9	VX	7	Virgin America	489
10	FL	7	AirTran Airways Corporation	263
11	YV	7	Mesa Airlines Inc.	81
12	AS	7	Alaska Airlines Inc.	62
13	F9	7	Frontier Airlines Inc.	58
14	HA	7	Hawaiian Airlines Inc.	31

Q9. Test your function to ensure it works. Create a new dataframe called sept that contains the results with only records from September and a dataframe called jan that contains the results with only records from January. The only change you should make to get the results is adjusting the value of the input parameter you are passing in the function call in num_flights_fun. Print out the last five

rows of sept and jan.

```
In [131... sept = num_flights_fun(9)
sept.tail()
```

```
Out[131...      carrier month      name num_flights
11      AS      9  Alaska Airlines Inc.      60
12      F9      9  Frontier Airlines Inc.      58
13      YV      9   Mesa Airlines Inc.      42
14      HA      9  Hawaiian Airlines Inc.      25
15      OO      9  SkyWest Airlines Inc.      20
```

```
In [132... jan = num_flights_fun(1)
jan.tail()
```

```
Out[132...      carrier month      name num_flights
11      AS      1  Alaska Airlines Inc.      62
12      F9      1  Frontier Airlines Inc.      59
13      YV      1   Mesa Airlines Inc.      46
14      HA      1  Hawaiian Airlines Inc.      31
15      OO      1  SkyWest Airlines Inc.       1
```

Q10. Using sept and jan, print the top five airlines with the greatest number of flights in September and in January, in descending order. Are the top five airlines the same in September and January? If so, what are the airlines? If not, which airlines differ from September to January?

```
In [133... top5_sept = (
    sept.sort_values("num_flights", ascending=False)
    .head(5)
)

top5_sept
```

```
Out[133...
```

	carrier	month	name	num_flights
0	EV	9	ExpressJet Airlines Inc.	4725
1	UA	9	United Air Lines Inc.	4694
2	B6	9	JetBlue Airways	4291
3	DL	9	Delta Air Lines Inc.	3883
4	AA	9	American Airlines Inc.	2614

```
In [134... top5_jan = (
    jan.sort_values("num_flights", ascending=False)
    .head(5)
)

top5_jan
```

```
Out[134...
```

	carrier	month	name	num_flights
0	UA	1	United Air Lines Inc.	4637
1	B6	1	JetBlue Airways	4427
2	EV	1	ExpressJet Airlines Inc.	4171
3	DL	1	Delta Air Lines Inc.	3690
4	AA	1	American Airlines Inc.	2794

```
In [135... sept_airlines = set(top5_sept["name"])
jan_airlines = set(top5_jan["name"])

sept_airlines == jan_airlines
```

```
Out[135... True
```

Airlines in the top five in September and January are United Air Lines, JetBlue Airways, ExpressJet Airlines, Delta Air Lines, American Airlines

Q11. Create a new function using pandas called `num_flights_fun_all` that runs the same query as above but returns the records for all months. The function should not take any arguments.

```
In [136... def num_flights_fun_all():
    """
    pandas.DataFrame
    Columns: carrier, month, name, num_flights
    """
```

```

query = """
SELECT
    f.carrier,
    f.month,
    a.name,
    count(*) AS num_flights
from
    flights f
join
    airlines a
    on f.carrier = a.carrier
group by f.carrier, f.month, a.name
order by f.month, num_flights DESC;
"""

df = pd.read_sql_query(query, conn)
return df

```

```

In [137... all_months = num_flights_fun_all()
all_months

```

```

Out[137...

```

	carrier	month	name	num_flights
0	UA	1	United Air Lines Inc.	4637
1	B6	1	JetBlue Airways	4427
2	EV	1	ExpressJet Airlines Inc.	4171
3	DL	1	Delta Air Lines Inc.	3690
4	AA	1	American Airlines Inc.	2794
...	...	...	...	...
180	FL	12	AirTran Airways Corporation	213
181	F9	12	Frontier Airlines Inc.	61
182	AS	12	Alaska Airlines Inc.	54
183	YV	12	Mesa Airlines Inc.	50
184	HA	12	Hawaiian Airlines Inc.	28

185 rows x 4 columns

Q12. Use `num_flights_fun_all` to create a dataframe called `months_df`. The dataframe should have `carrier`, `month`, `name`, and `num_flights` columns and contain the number of flights that each airline made in each month. Show the first 15 lines of the dataframe to ensure the results are correct.

```
In [138... months_df = num_flights_fun_all()
months_df.head(15)
```

```
Out[138...   carrier  month                name  num_flights
0      UA      1    United Air Lines Inc.         4637
1      B6      1      JetBlue Airways         4427
2      EV      1    ExpressJet Airlines Inc.         4171
3      DL      1      Delta Air Lines Inc.         3690
4      AA      1    American Airlines Inc.         2794
5      MQ      1        Envoy Air         2271
6      US      1      US Airways Inc.         1602
7      9E      1    Endeavor Air Inc.         1573
8      WN      1    Southwest Airlines Co.          996
9      FL      1  AirTran Airways Corporation         328
10     VX      1      Virgin America         316
11     AS      1    Alaska Airlines Inc.          62
12     F9      1    Frontier Airlines Inc.          59
13     YV      1    Mesa Airlines Inc.          46
14     HA      1    Hawaiian Airlines Inc.          31
```

Q13. Use `months_df` to plot the number of flights each airline made for each month using `altair`, `plotly`, or `seaborn`.

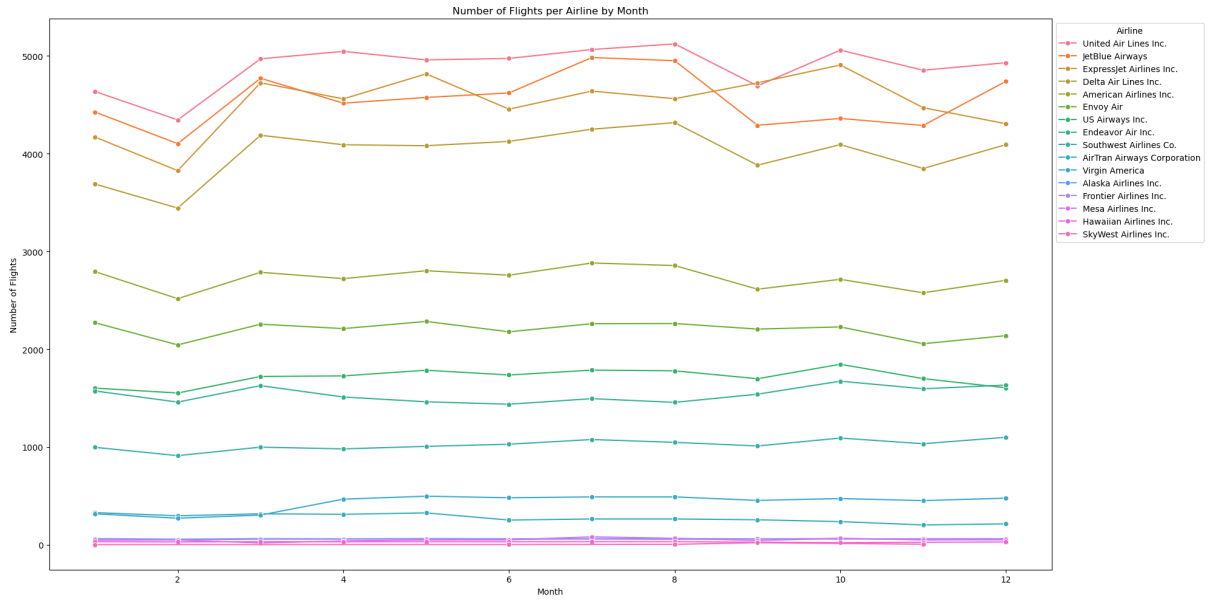
```
In [139... import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [140... plt.figure(figsize=(20, 10))

sns.lineplot(
    data=months_df,
    x="month",
    y="num_flights",
    hue="name",
    marker="o"
)

plt.title("Number of Flights per Airline by Month")
plt.xlabel("Month")
plt.ylabel("Number of Flights")
plt.legend(title="Airline", bbox_to_anchor=(1, 1), loc="upper left")
```

```
plt.tight_layout()
plt.show()
```



Q14. Write a query that returns the each airlines' mean arrival delay across all of its flights. The result should have two columns: mean_delay and the full name of the airline. Which airline has the highest average delay?

```
In [141]: %sql
select
    a.name,
    round(avg(f.arr_delay), 2) as mean_delay
from
    flights f
join
    airlines a
    on f.carrier = a.carrier
where f.arr_delay is not null
group by a.name
order by mean_delay DESC

* sqlite:///airline_industry.db
Done.
```

Out [141...

name	mean_delay
Frontier Airlines Inc.	21.79
AirTran Airways Corporation	19.59
ExpressJet Airlines Inc.	14.9
Mesa Airlines Inc.	14.08
SkyWest Airlines Inc.	10.81
Envoy Air	10.22
Southwest Airlines Co.	9.47
JetBlue Airways	9.36
Endeavor Air Inc.	6.91
United Air Lines Inc.	3.5
US Airways Inc.	2.06
Virgin America	1.75
Delta Air Lines Inc.	1.63
American Airlines Inc.	0.36
Hawaiian Airlines Inc.	-6.92
Alaska Airlines Inc.	-9.86

In [142...

```
%%sql
select
    a.name,
    round(avg(f.arr_delay), 2) as mean_delay
from
    flights f
join
    airlines a
    on f.carrier = a.carrier
where f.arr_delay is not null
group by a.name
order by mean_delay DESC
limit 1
```

```
* sqlite:///airline_industry.db
```

Done.

Out [142...

name	mean_delay
Frontier Airlines Inc.	21.79

Q15. Use pandas to query the data and return a dataframe to plot

the data. Use either altair, seaborn, or plotly and plot the mean delay for each airline in a bar plot. The plot should have the airline name on the y and mean delay on the x and be ordered by mean delay. The bars should be colored by whether the airlines' flights arrive, on average, early or late. Which airline, on average, has the earliest arrivals?

```
In [143... import pandas as pd

query = """
select
    a.name as airline_name,
    avg(f.arr_delay) as mean_delay
from flights f
join airlines a
    on f.carrier = a.carrier
where f.arr_delay is not null
group by a.name
"""

delay_df = pd.read_sql_query(query, conn)

delay_df["mean_delay"] = delay_df["mean_delay"].round(2)

delay_df["status"] = delay_df["mean_delay"].apply(lambda x: "Early (avg < 0)" if x < 0 else "Late (avg > 0)")

delay_df = delay_df.sort_values("mean_delay", ascending=True)

delay_df.head()
```

```
Out[143...      airline_name  mean_delay  status
1  Alaska Airlines Inc.    -9.86  Early (avg < 0)
8  Hawaiian Airlines Inc.   -6.92  Early (avg < 0)
2  American Airlines Inc.    0.36  Late (avg > 0)
3   Delta Air Lines Inc.    1.63  Late (avg > 0)
15   Virgin America      1.75  Late (avg > 0)
```

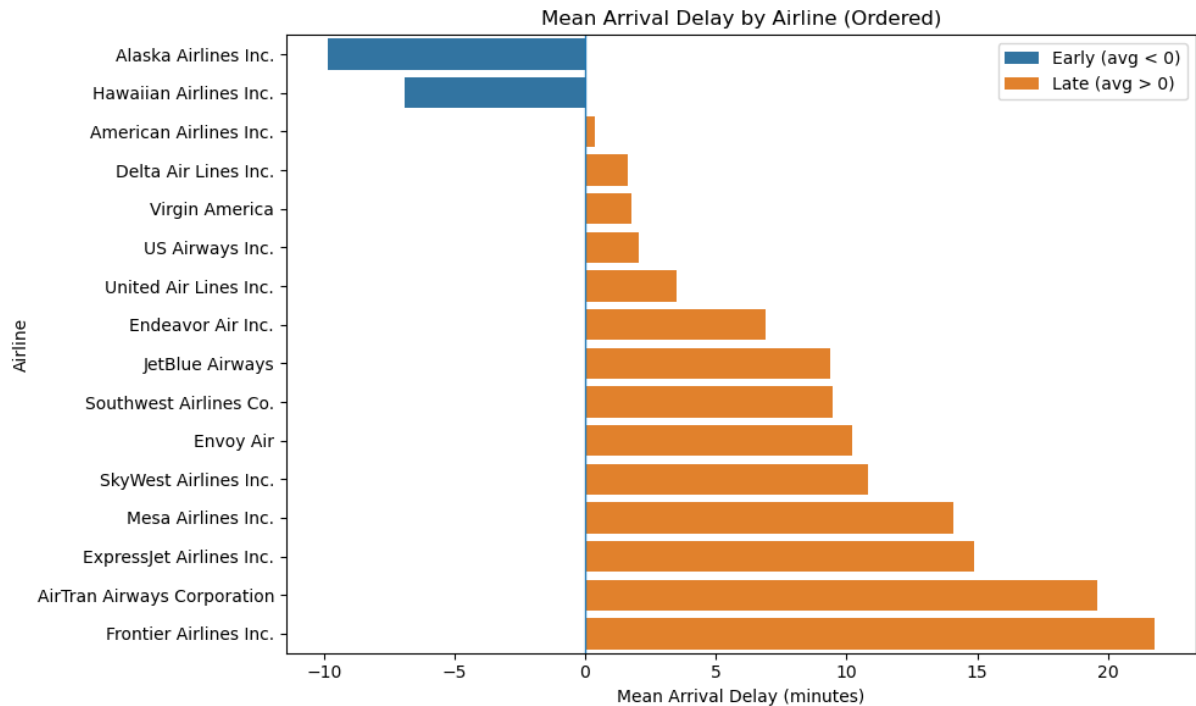
```
In [144... import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))

sns.barplot(
    data=delay_df,
    y="airline_name",
    x="mean_delay",
    hue="status",
    dodge=False
```

```
)

plt.title("Mean Arrival Delay by Airline (Ordered)")
plt.xlabel("Mean Arrival Delay (minutes)")
plt.ylabel("Airline")
plt.axvline(0, linewidth=1)
plt.legend(title="")
plt.tight_layout()
plt.show()
```



```
In [145... earliest_airline = delay_df.iloc[0][["airline_name", "mean_delay"]]
earliest_airline
```

```
Out[145... airline_name    Alaska Airlines Inc.
mean_delay              -9.86
Name: 1, dtype: object
```